

LABORATORY PROJECT REPORT
CONTROLLING A DC MOTOR USING L298P MOTOR
SHIELD AND GPIO
EXPERIMENT 5

Section 1

Group 2

Semester 1 2025/2026

NO	NAME	MATRIC NO
1.	ADAM NURUDDIN BIN HELMI	2215783
2.	NUR FATIHAH HANNANI BINTI HASMAYADI	2227062
3.	NUR SYAHZMAN AZIQ BIN MOHD SHAHMADI	2310037

Date of submission: 12th November 2025

Abstract

This experiment focuses on controlling the speed and direction of a DC motor using the LSS05 Motor Driver Shield and an Arduino Uno microcontroller. The LSS05 shield functions similarly to the L298P H-Bridge driver but provides more efficient control with simplified pin mapping and onboard power management. The experiment aims to demonstrate how Pulse Width Modulation (PWM) signals can be used to vary motor speed and how digital pins determine rotational direction. By programming the Arduino through the Arduino IDE, the team successfully implemented motor control in both forward and reverse directions while observing the relationship between PWM duty cycle and motor performance. The experiment enhances students' understanding of DC motor interfacing and practical speed control methods used in mechatronic systems.

Table of Contents

1.0 Introduction

2.0 Materials and Equipment

2.1 Electronic Components

2.2 Equipment and Tools

3.0 Experimental Setup

4.0 Methodology

5.0 Data Collection

6.0 Data Analysis

7.0 Results

8.0 Discussion

9.0 Conclusion

10.0 Recommendations

11.0 References

1.0 Introduction

In mechatronic systems, DC motors are essential components that convert electrical energy into mechanical motion, enabling movement in robotic and automation applications. To effectively control a DC motor, it is necessary to manage both the direction and speed of rotation. This can be achieved using a motor driver shield, which acts as an interface between the low-power control signals from the Arduino and the higher current required by the motor.

The LSS05 Motor Driver Shield used in this experiment integrates two H-bridge circuits, allowing bidirectional control of two DC motors. It uses GPIO pins to set motor direction and PWM signals to control motor speed. PWM varies the effective voltage applied to the motor by adjusting the signal's duty cycle—higher duty cycles produce faster rotation, while lower ones slow it down.

This experiment aims to familiarize students with the principles of DC motor control, including the implementation of PWM for speed regulation and the practical use of the LSS05 shield with Arduino. By conducting tests at different PWM values and observing motor behavior, students gain insight into the interaction between hardware and software in motion control applications.

2.0 Materials and Equipment

2.1 Electronic Components

- Arduino Board – Microcontroller board for controlling the display
- LSS05 Motor Driver Shield - to read data such as detecting hand gestures
- DC Motor - establish connections between components

2.2 Equipment and Tools

- Arduino IDE – Software for writing and uploading code to the Arduino Uno
- USB cable – For connecting the Arduino Uno to a computer
- Power Supply (5V via USB or external source)
- Motor Driver Power supply (Optional)

3.0 Experimental Setup

1. Attached the LSS05 motor driver shield onto the arduino uno R3.
2. Make sure the power source selection jumper is to 5v arduino power.
3. Connect the Motor A to motor terminal 1, and Motor B to motor terminal 2.
4. Make sure the wire of motor A and B were screwed correctly to the motor terminals.



4.0 Methodology

1. Setup the Arduino UNO R3
2. Setup the Motor Driver Shield
3. IDE code implementation
4. Experiment the PWM value of DC Motor
5. Observe the DC motor speed change, and the rotation.
6. Code snippet

4.1 Detailed steps

- 1) Assemble the circuit as stated in the experimental setup.
- 2) Connect the arduino to the computer via usb cable, and upload the code to arduino IDE.
- 3) Adjust the arduino IDE code with the respective PWM value and rotation of DC motor.
- 4) Observe the change of the speed of DC motor, and estimate it's RPM.

4.2 Programming codes

Arduino Codes

```
// Motor A pins (based on LSS05 shield)

const int ENA = 5; // PWM for Motor A

const int IN1 = 4; // Direction

const int IN2 = 7; // Direction


// Motor B pins (optional)

const int ENB = 6; // PWM for Motor B

const int IN3 = 8; // Direction

const int IN4 = 9; // Direction

const int spdA =200;
```

```
const int spdB =200;

void setup() {

  pinMode(ENA, OUTPUT);

  pinMode(IN1, OUTPUT);

  pinMode(IN2, OUTPUT);

  pinMode(ENB, OUTPUT);

  pinMode(IN3, OUTPUT);

  pinMode(IN4, OUTPUT);

}


void loop() {

  // --- Motor A forward ---

  digitalWrite(IN1, HIGH);

  digitalWrite(IN2, LOW);

  analogWrite(ENA, spdA);


  // --- Motor B forward (optional) ---

  digitalWrite(IN3, HIGH);

  digitalWrite(IN4, LOW);

  analogWrite(ENB, 128);

  delay(2000);


  // --- Motor A backward ---
```

```

digitalWrite(IN1, LOW);

digitalWrite(IN2, HIGH);

analogWrite(ENA, 64);

// --- Motor B backward (optional) ---

digitalWrite(IN3, LOW);

digitalWrite(IN4, HIGH);

analogWrite(ENB, spdB);

delay(2000);

// --- Stop all motors ---

analogWrite(ENA, 0);

analogWrite(ENB, 0);

delay(500);

}

```

5.0 Data Collection

PWM VALUE	DIRECTION	OBSERVED SPEED (RPM est.)
255	Forward	228
128	Forward	142
255	Reverse	228
64	Reverse	100

6.0 Data Analysis

A definite proportionate trend can be seen in the link between the PWM value and the observed motor speed based on the data gathered. The motor speed rises in proportion to the PWM value. In the forward motion, a maximum PWM value of 255 generated a speed of 228 RPM, whilst a PWM value of 128 produced a speed of 142 RPM. In the other way, a maximum PWM value of 255 produced 228 RPM, while a lower PWM value of 64 produced 100 RPM. Because greater PWM values result in a higher average voltage, which accelerates the motor, this shows that the PWM signal efficiently regulates the motor speed. Additionally, when the same PWM value is applied, the motor speed is almost the same in both directions, demonstrating that the direction control circuit functions correctly and only modifies the rotation direction rather than the speed magnitude. Any little changes in speed might be caused by outside variables like friction, voltage swings, or inaccurate sensors.

7.0 Results

The findings demonstrate that the motor speed rises in direct proportion to the PWM input value. For both forward and reverse directions, the maximum speed was 228 RPM at a PWM value of 255, while the lowest speed was 100 RPM at a PWM value of 64. The control system's ability to precisely regulate speed and direction was confirmed by the motor's constant performance in both directions. All things considered, the experiment effectively shows that PWM control offers a dependable and effective way to modify DC motor speed.

8.0 Discussion

This experiment showed that changing the PWM value has a clear and direct effect on the speed of the DC motor. When the PWM value was high, the motor spin faster, and when the PWM value was low, the motor slowed down. For example, the highest PWM value of 255 produced the top speed of 228 RPM while lower values like 128 and 64 produced slower speeds. The higher PWM value means the motor receives more average voltage causing it to rotate more faster.

The results also showed that the motor behaves almost the same in both forward and reverse directions when using the same PWM value. This means the motor driver was working correctly because it only changed the motor's direction and did not affect the power or speed. Any small differences in speed were likely caused by normal factors such as friction, slight changes in the battery voltage, or small errors in measurement.

Overall, the experiment proved that PWM is a simple and effective way to control the speed of a DC motor. It allowed the smoothness of speed changes and reliable direction control. These results show that PWM control is useful for many applications such as robots, automatic systems, and small motor-based machines where speed needs to be adjusted accurately and easily.

9.0 Conclusion

In conclusion, this experiment successfully showed how PWM can be used to control the speed and direction of a DC motor. By adjusting the PWM values, the motor speed increased or decreased in a predictable way. It confirming that PWM is effective method for motor speed control. The motor also worked smoothly in both forward and reverse directions, proving that the driver circuit correctly handled direction changes without affecting performance.

10.0 Recommendation

Our recommendation is to get enough supply to the motor with a stable external power source instead of relying on USB power from the Arduino. A separate power supply can help avoid voltage drops that may affect the motor's performance. With a separate power source, the motor receives a stable voltage, which helps maintain consistent speed when different PWM values are applied.

12.0 Acknowledgement

Special thanks to ZULKIFLI BIN ZAINAL ABIDIN & WAHJU SEDIONO for their guidance and support during this experiment.

Student's Declaration

Certificate of Originality and Authenticity

We hereby certify that we are collectively responsible for the work presented in this report. The content reflects our original work, except where specific references or acknowledgements have been made. No part of this report has been completed or submitted by individuals or sources not identified within.

Furthermore, we confirm that this report is the result of a collaborative effort, with contributions made by all group members. The extent of each individual's contribution is documented within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

Name: ADAM NURUDDIN BIN HELMI	Read	[/]
Matric Number: 2215783	Understand	[/]
Signature: <i>ADAM</i>	Agree	[/]

Name: NUR FATIHAH HANNANI BINTI HASMAYADI	Read	[/]
Matric Number: 2227062	Understand	[/]
Signature: <i>FAT</i>	Agree	[/]

Name: NUR SYAHZMAN AZIQ BIN MOHD SHAHMADI	Read	[/]
Matric Number: 2310037	Understand	[/]
Signature: <i>AZIQ</i>	Agree	[/]